

Question	Marking Guidance	Mark	Comments
06.1	Correct answer 23.55 – 24 two marks;; For one mark 5.9 OR 94.2;	2	
06.2	1. Method for measuring area; eg draw round (each) leaf on graph paper and count squares; 2. Of both sides of (each) leaf; 3. Divide rate (of water loss/uptake from potometer) by (total) surface area (of leaves);	3	
06.3	Plant has roots OR xylem cells very narrow;	1	Ignore references to air bubbles/mass flow/photosynthesis Accept xylem damaged when cut
06.4	1. Both small/similar size (so fit channel); 2. Have a similar shape (so bind to/fit channel);	2	1. Accept same height and width Ignore refs to polar/non-polar 2. Accept Aquaporin complementary to oxygen(s)
06.5	1. Single-stranded RNA (has base sequence) complementary to PIP1 mRNA; 2. Binds to mRNA (of PIP1)/leads to destruction of mRNA; 3. Prevents/reduces translation (of <i>PIP1</i>); 4. Reduces photosynthesis/named process that uses water;	3 max	3. Less made is insufficient

06.6	Not all of mRNA bound to single-stranded RNA/there is more mRNA than interfering RNA OR Not all mRNA destroyed/disabled;	1	Accept mutations in transgene, Accept not all cells with transgenes
06.7	1. Loss of PIP reduces water and carbon dioxide movement; 2. Differences significant because SDs don't overlap OR Need stats test to see whether significant differences (or not); 3. Greater (proportional) effect on carbon dioxide transport; 4. Not all movement through PIP;	3 max	1. Accept converse for wild type 2. Reject references to results significant or not significant 2. Accept error bars for SDs